Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

Special Examination

July 2025

Max. Marks: 100 | **Duration: 3 Hrs**

Class: TE | **Semester: V**

Course Code: CS301 | **Program: CE/CSE-AIML/CSE-DS**

Name of the Course: Theory of Computation

Instruction:

(1) All questions are compulsory

(2) Partial credit will be given, so show your work where appropriate

(3) Note: DFA = Deterministic Finite Automata, NFA = Nondeterministic Finite Automata, PDA = Push-Down Automata, CFG=Context Free Grammar, TM = Turing Machine.

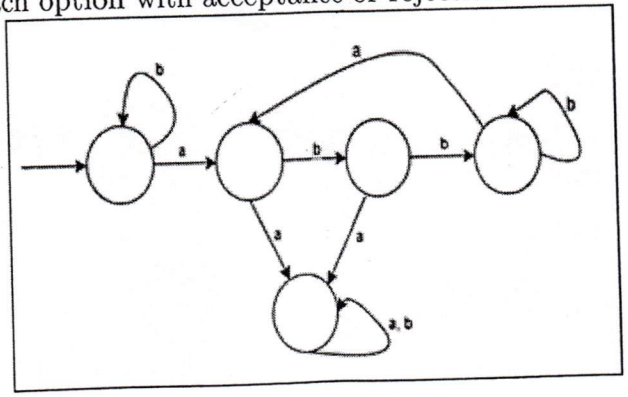
Course Outcomes At the end of the course student will be able to:

(1) Design finite automaton for regular expressions and languages.

(2) Apply the properties of regular languages.

(3) Construct the grammar for a language and convert it into normal forms.

(4) Design and evaluate pushdown automata and Turing Machine for a language.

Q No.		Max. Marks	CO-BL
Q.1a	Design a minimal DFA that accepts all strings over the alphabet (a, b) where every accepted string w starts with substring s= "abb". Validate the machine based on the user's input.	10	1-3
Q.1b	Consider the machine M below. The Language accepted by the machine is: a. $\{w \in \{a,b\}^{\star}$: every a in w is followed by exactly two b's $\}$ b. $\{w \in \{a,b\}^{\star}$: every a in w is followed by at least two b's $\}$ c. $\{w \in \{a,b\}^{\star} : w$ contains the substring $abb\}$ d. $\{w \in \{a,b\}^{\star} : w$ does not contain aa as a substring $\}$ Justify each option with acceptance or rejection. 	10	2-3

Q.2a	Convert the following grammar to Greibach normal forms (GNF). $S \rightarrow XB \mid AA$ $A \rightarrow a \mid SA$ $B \rightarrow b$ $X \rightarrow a$	10	3-3

Q.2b	Given the following state table of an FSM with two states A and B, one input and one output: if the initial state is $A = 0, B = 0$, what is the minimum length of an input string which will take the machine to the state $A = 0, B = 1$ with output=1?	10	1-3

Present state A	Present state B	input	Next state A	Next state B	output
0	0	0	0	0	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	1	0	0
0	0	1	0	1	0
0	1	1	0	0	1
1	0	1	0	1	1
1	1	1	0	0	1

Q.3a	Consider the grammar consisting of 7 productions: $S \rightarrow aA \mid aBB$ $A \rightarrow aaA \mid \epsilon$ $B \rightarrow bB \mid bbC$ $C \rightarrow B$ Simplify the given grammar, how may productions remain in the simplified grammar? **OR** Convert the given grammar into Chomsky normal form (CNF): $S \rightarrow aAb \mid bB$ $A \rightarrow a \mid b$ $B \rightarrow b$	10	3-3
Q.3b	Show that the following grammar is ambiguous with respect to the string *aaabbabbba*. $S \rightarrow aB \mid bA$ $A \rightarrow aS \mid bAA \mid a$ $B \rightarrow bS \mid aBB \mid b$	10	3-3
Q.4a	Design a Turing Machine to accept the following language $L = \{0^n 1^n \mid n \geq 1\}$. Also state with proof whether L is a regular language or not. **OR** Prove that there are languages which can be accepted by a Turing Machine but not by a push down automata.	10	4-3
Q.4b	Give an example of a language which can be accepted by a non-deterministic push down automata(NPDA) but not by a deterministic push down automata (DPDA). Construct the corresponding NPDA. Also state briefly why your chosen language cannot be accepted by a DPDA.	10	4-3
Q.5a	Give an example of an undecidable language. Also give a proof why it is undecidable.	10	4-3
Q.5b	Define recursive enumerable and recursive languages. Are there languages which are recursively enumerable but not recursive? Explain with proof.	10	4-3